

gilmore-lie-groups: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_TAB_001	26	0.877	$\begin{tabular}{t}{ l l l l l }\hline & I & (123) & (321) & \text{Basis functions} \\ \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{tabular}$	$\begin{array}{c} \Gamma^1 \\ \Gamma^3 \end{array} \left \begin{array}{ccc} 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{array} \right. \begin{array}{l} v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ v_3 = s_1 + \omega^2 s_2 + \omega s_3 \end{array}$	
gilmore-lie-groups_TAB_002	30	0.669	$\begin{tabular}{t}{ l l l l l l }\hline & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{tabular}$	$\begin{array}{c} \Gamma^1 \\ \Gamma^2 \\ \Gamma^3 \\ \Gamma^4 \end{array} \left \begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array} \right. \begin{array}{l} w_1 = t_1 + t_2 + t_3 + t_4 \\ w_2 = t_1 + t_2 - t_3 - t_4 \\ w_3 = t_1 - t_2 + t_3 - t_4 \\ w_4 = t_1 - t_2 - t_3 + t_4 \end{array}$ The linear combinations of these roots that transform under each of the	
gilmore-lie-groups_TAB_003	62	0.837	$\begin{tabular}{t}{ l l }\hline & \text{Group} & \text{Dimension} \\ \hline (U T(p, q)) & (p^2 + q^2 + p q) \\ (H T(p, q)) & p(p + q) \\ (U T(p, q, r)) & p^2 + q^2 + r^2 + p q + p r + q r \\ (Sol(n)) & n(n + 1)/2 \\ (Nil(n)) & n(n - 1)/2 \\ (A(p, q)) & p q \\ \hline\end{tabular}$	$\begin{array}{c} \text{Group} \\ UT(p, q) \\ HT(p, q) \\ UT(p, q, r) \\ Sol(n) \\ Nil(n) \\ A(p, q) \end{array} \begin{array}{c} \text{Dimension} \\ p^2 + q^2 + pq \\ p(p + q) \\ p^2 + q^2 + r^2 + pq + pr + qr \\ n(n + 1)/2 \\ n(n - 1)/2 \\ pq \end{array}$	
gilmore-lie-groups_TAB_004	62	1.000	$\begin{tabular}{t}{ l l l }\hline & \text{Transformation} & \text{Invariant} \\ \hline Galilean & invariant \\ Poincaré & invariant \end{tabular}$	$\begin{array}{c} \text{Transformation} \\ \text{Galilean} \\ \text{Poincaré} \end{array} \begin{array}{c} \mathbf{F} = d\mathbf{p}/dt \\ \text{invariant} \\ \text{not invariant} \end{array} \begin{array}{c} \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) A_\mu = -\frac{4\pi}{c} j_\mu \\ \text{not invariant} \\ \text{invariant} \end{array}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ TAB_009	151	■ 1.000	<pre> \begin{tabular}[t]{ l l l } \hline \(\left[\begin{array}{cc c}0 & \theta & v_{1}\ \ -\theta & 0 & v_{2}\ \ \hline v_{1} & v_{2} & 0\end{array}\right]\) & \(\left[\begin{array}{cc cc}0 & \theta & v_{1} & t_{1} \\ -\theta & 0 & v_{2} & t_{2} \\ \hline v_{1} & v_{2} & 0 & t_{3} \\ 0 & 0 & 0 & 0\end{array}\right]\) & \(\left[\begin{array}{cc cc}0 & \theta & v_{1} & t_{1} \\ -\theta & 0 & v_{2} & t_{2} \\ \hline 0 & 0 & 0 & t_{3} \\ 0 & 0 & 0 & 0\end{array}\right]\) \\ \hline Lorentz & Poincare & Galilei \\ \hline \end{tabular} </pre>	$\left[\begin{array}{cc c}0 & \theta & v_{1} \\ -\theta & 0 & v_{2} \\ \hline v_{1} & v_{2} & 0\end{array}\right]$ $\left[\begin{array}{cc cc}0 & \theta & v_{1} & t_{1} \\ -\theta & 0 & v_{2} & t_{2} \\ \hline v_{1} & v_{2} & 0 & t_{3} \\ 0 & 0 & 0 & 0\end{array}\right]$ $\left[\begin{array}{cc cc}0 & \theta & v_{1} & t_{1} \\ -\theta & 0 & v_{2} & t_{2} \\ \hline 0 & 0 & 0 & t_{3} \\ 0 & 0 & 0 & 0\end{array}\right]$ Lorentz Poincare Galilei	

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gilmore-lie-groups_ TAB_010	167	0.608	$\begin{array}{l} \begin{array}{l} \hline \left[H_1, H_2 \right] \&= \& 0 \\ \hline \left[\mathbf{H}, E_{\pm 2\mathbf{e}_1} \right] \&= \& \left(\pm 2/\sqrt{12}, 0 \right) E_{\pm 2\mathbf{e}_1} \\ \hline \left[\mathbf{H}, E_{\pm 2\mathbf{e}_2} \right] \&= \& \left(0, \pm 2/\sqrt{12} \right) E_{\pm 2\mathbf{e}_2} \\ \hline \left[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \right] \&= \& \left(\pm/\sqrt{12}, \pm/\sqrt{12} \right) E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ \hline \left[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1} \right] \&= \& (2/\sqrt{12}) H_1 \\ \hline \left[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2} \right] \&= \& (2/\sqrt{12}) H_2 \\ \hline \left[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)} \right] \&= \& (1/\sqrt{12}) (\pm H_1 \pm H_2) \\ \hline \left[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)} \right] \&= \& *(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2} \\ \hline \left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2} \right] \&= \& (1/\sqrt{6}) E_{-2\mathbf{e}_1} \\ \hline \left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1} \right] \&= \& (1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \hline \left[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2} \right] \&= \& (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ \hline \left[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2} \right] \&= \& (-1/\sqrt{6}) E_{+2\mathbf{e}_1} \\ \hline \left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2} \right] \&= \& (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \hline \left[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2} \right] \&= \& *(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ \hline \left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2} \right] \&= \& (1/\sqrt{6}) E_{-2\mathbf{e}_2} \\ \hline \left[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2} \right] \&= \& (1/\sqrt{6}) E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ \hline \left[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2} \right] \&= \& (-1/\sqrt{6}) E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \hline \left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2} \right] \&= \& (-1/\sqrt{6}) E_{+2\mathbf{e}_2} \\ \hline \left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2} \right] \&= \& (-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{array} \end{array}$	<table><tr><td>$[H_1, H_2]$</td><td>=</td><td>0</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$</td><td>=</td><td>$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$</td><td>=</td><td>$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$</td><td>=</td><td>$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$</td></tr><tr><td>$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$</td><td>=</td><td>$(2/\sqrt{12})H_1$</td></tr><tr><td>$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(2/\sqrt{12})H_2$</td></tr><tr><td>$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$</td><td>=</td><td>$(1/\sqrt{12})(\pm H_1 \pm H_2)$</td></tr><tr><td>$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$</td><td>=</td><td>$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{-2\mathbf{e}_1}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$</td><td>=</td><td>$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{-2\mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$</td></tr><tr><td>$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$</td></tr></table>	$[H_1, H_2]$	=	0	$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$	=	$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$	=	$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$	$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$	=	$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$	$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$	=	$(2/\sqrt{12})H_1$	$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(2/\sqrt{12})H_2$	$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	=	$(1/\sqrt{12})(\pm H_1 \pm H_2)$	$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	=	$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$	$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$	9.8 Conclusion
$[H_1, H_2]$	=	0																																																												
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gilmore-lie-groups_ TAB_011	172	■1.000	<pre>\begin{tabular}[t]{l l l l l l} \hline \multirow[b]{3}{*}{\left((i \ \phi)^{\wedge}{j} \ f_{\jmath}\right)} & & & & & \\ \multicolumn{4}{ c }{Representation} \ \ \ \hline & \left(\frac{1}{2}\right) & 1 & \left(\frac{3}{2}\right) & 2 \ \ \hline & 3 & 4 & 5 \ \ \hline (f_{\scriptscriptstyle 0}) & \cos(\phi/2) & 1 & \frac{9}{8}\cos(\frac{\phi}{2})-\frac{1}{8}\cos(\frac{3\phi}{2}) & 1 \\ (i \ \phi)^1 f_1 & 2i \sin(\phi/2) & i \sin(\phi) & \frac{9i}{4}\sin(\frac{\phi}{2})-\frac{i}{12}\sin(\frac{3\phi}{2}) & \frac{i}{3}\sin(\phi)-\frac{i}{6}\sin(2\phi) \\ (i \ \phi)^2 f_2 & & \cos(\phi)-1 & -\frac{1}{2}\cos(\frac{\phi}{2})+\frac{1}{2}\cos(\frac{3\phi}{2}) & -\frac{5}{4}+\frac{1}{3}\cos(\phi)-\frac{1}{12}\cos(2\phi) \\ (i \ \phi)^3 f_3 & & & -i \sin(\frac{\phi}{2})+\frac{i}{3}\sin(\frac{3\phi}{2}) & -\frac{i}{3}\sin(\phi)+\frac{i}{6}\sin(2\phi) \\ (i \ \phi)^4 f_4 & & & & \frac{1}{4}-\frac{1}{3}\cos(\phi)+\frac{1}{12}\cos(2\phi) \\ \hline \hline \end{tabular}</pre>	(not rendered)	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:																																								
gilmore-lie-groups_ TAB_012	175	■0.995	<pre>\begin{tabular}[t]{l l l l l l} \hline \left(\cos^{\wedge}{2}(\alpha, \beta)\right) & \left(\theta(\alpha, \beta)\right) & n=\frac{2\alpha\cdot\beta}{\alpha\cdot\alpha} & n'=\frac{2\alpha\cdot\beta}{\beta\cdot\beta} & \frac{\alpha\cdot\alpha}{\beta\cdot\beta}=\frac{n'}{n} \\ 1 & \frac{\pi}{2}\pm\frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ \frac{3}{4} & \frac{\pi}{2}\pm\frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{2}{4} & \frac{\pi}{2}\pm\frac{\pi}{4} & \pm 2 & \pm 3 & 3^{+1} \\ \frac{1}{4} & \frac{\pi}{2}\pm\frac{\pi}{6} & \pm 1 & \pm 2 & 2^{-1} \\ 0 & \frac{\pi}{2} & 0 & 0 & 2^{+1} \\ \hline \hline \end{tabular}</pre>	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>3</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td></td><td></td><td>± 1</td><td>± 3</td><td>3^{+1}</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td></td><td></td><td>± 1</td><td>± 2</td><td>2^{+1}</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>-</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	3	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}			± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}			± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	-	
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1																																									
3	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}																																									
		± 1	± 3	3^{+1}																																									
$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}																																									
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$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1																																									
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gilmore-lie-groups_ TAB_013	180	0.508	<pre>\begin{tabular}[t]{ l l l l l l } \hline & \alpha_1 & \alpha_2 & \alpha_3 & \dots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \dots & \mathbf{e}_{l-1} - \mathbf{e}_l & \mathbf{e}_{l-1} + \mathbf{e}_l \\ D_l & \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 & \dots & \mathbf{e}_l & \mathbf{e}_l \\ B_l & \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 & \dots & \mathbf{e}_l & \mathbf{e}_l \\ C_l & \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 & \dots & \mathbf{e}_l & 2\mathbf{e}_l \end{tabular}</pre>	<table><tr><td></td><td>α_1</td><td>α_2</td><td>α_3</td><td></td><td>α_{l-1}</td><td>α_l</td></tr><tr><td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>...</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td><td></td></tr><tr><td>D_l</td><td>$\mathbf{e}_1$</td><td>$\mathbf{e}_2$</td><td>$\mathbf{e}_3$</td><td>...</td><td>$\mathbf{e}_l$</td><td>$\mathbf{e}_{l-1} + \mathbf{e}_l$</td></tr><tr><td>$B_l$</td><td>$\mathbf{e}_1$</td><td>$\mathbf{e}_2$</td><td>$\mathbf{e}_3$</td><td>...</td><td>$\mathbf{e}_l$</td><td>$\mathbf{e}_l$</td></tr><tr><td>$C_l$</td><td>$\mathbf{e}_1$</td><td>$\mathbf{e}_2$</td><td>$\mathbf{e}_3$</td><td>...</td><td>$\mathbf{e}_l$</td><td>$2\mathbf{e}_l$</td></tr></table>		α_1	α_2	α_3		α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	...	$\mathbf{e}_{l-1} - \mathbf{e}_l$		D_l	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_3$	...	$\mathbf{e}_l$	$\mathbf{e}_{l-1} + \mathbf{e}_l$	B_l	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_3$	...	$\mathbf{e}_l$	$\mathbf{e}_l$	C_l	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_3$	...	$\mathbf{e}_l$	$2\mathbf{e}_l$	$A_{l-1}$$\mathbf{e}_1 - \mathbf{e}_2$$\mathbf{e}_2 - \mathbf{e}_3$$\mathbf{e}_3 - \mathbf{e}_4$$\dots$$\mathbf{e}_{l-1} - \mathbf{e}_l$ $D_l$$\mathbf{e}_1$$\mathbf{e}_2$$\mathbf{e}_3$$\dots$$\mathbf{e}_l$$\mathbf{e}_{l-1} + \mathbf{e}_l$ $B_l$$\mathbf{e}_1$$\mathbf{e}_2$$\mathbf{e}_3$$\dots$$\mathbf{e}_l$$\mathbf{e}_l$ $C_l$$\mathbf{e}_1$$\mathbf{e}_2$$\mathbf{e}_3$$\dots$$\mathbf{e}_l$$2\mathbf{e}_l$ Inner products among the fundamental roots are summarized conv
	α_1	α_2	α_3		α_{l-1}	α_l																																		
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gilmore-lie-groups_ TAB_014	182	0.999	<pre>\begin{tabular}[t]{ l l l l } \hline p & q & r & Root space \\ p & 2 & 2 & D_{p+2} \\ 3 & 3 & 2 & E_6 \\ 4 & 3 & 2 & E_7 \\ 5 & 3 & 2 & E_8 \end{tabular}</pre>	<table><tr><td>p</td><td>q</td><td>r</td><td>Root space</td></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8	$3$$3$$2$$E_6$ $4$$3$$2$$E_7$ $5$$3$$2$$E_8$ grams are summarized in 7 : root spaces. Each root sp															
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gilmore-lie-groups_ TAB_015	184	1.000	<pre>\begin{tabular}[t]{ l l l l } \hline Canonical form & Boson operators & Coordinates and derivatives & Fermion operators \\ H_i & b_i^\dagger b_i & x^i \partial_i & f_i^\dagger f_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j & f_i^\dagger f_j \end{tabular}</pre>	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$H_i$$b_i^\dagger b_i$$x^i \partial_i$$f_i^\dagger f_i$ $E_{+\mathbf{e}_i - \mathbf{e}_j}$$b_i^\dagger b_j$$x^i \partial_j$$f_i^\dagger f_j$ that the following identifications are appropriate for the eigenopera paces C_l and D_l:																							
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																																										
gilmore-lie-groups_ TAB_018	195	■ 0.500	$\begin{tabular}[t]{ l l l l l } \hline (A_1) & \sim & (B_1) & \sim & (C_1) \\ \hline \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \hline \mathfrak{su}(1,1) = \mathfrak{sl}(2;\mathbb{R}) & \sim & \mathfrak{so}(2,1) & \sim & \mathfrak{sp}(2;\mathbb{R}) \\ \hline D_2 & = & A_1 & + & A_1 \\ \hline \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \hline \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \hline \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2;\mathbb{C}) & & \\ \hline \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ \hline B_2 & = & C_2 & & \\ \hline \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \hline \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \hline \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4;R) & & \\ \hline D_3 & = & A_3 & & \\ \hline \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \hline \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \hline \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \hline \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \hline \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4;\mathbb{R}) & & \\ \hline \hline \end{tabular}$	<table><tr><td>A_1</td><td></td><td>B_1</td><td></td><td>C_1</td></tr><tr><td>$\mathfrak{su}(2)$</td><td></td><td>$\mathfrak{so}(3)$</td><td></td><td>$\mathfrak{sp}(1) = \mathfrak{usp}(2)$</td></tr><tr><td>$\mathfrak{su}(1,1) = \mathfrak{sl}(2;\mathbb{R})$</td><td></td><td>$\mathfrak{so}(2,1)$</td><td></td><td>$\mathfrak{sp}(2;\mathbb{R})$</td></tr><tr><td>$D_2$</td><td>=</td><td>$A_1$</td><td>+</td><td>$A_1$</td></tr><tr><td>$\mathfrak{so}(4)$</td><td>=</td><td>$\mathfrak{so}(3)$</td><td>+</td><td>$\mathfrak{so}(3)$</td></tr><tr><td>$\mathfrak{so}^*(4)$</td><td></td><td>$\mathfrak{so}(3)$</td><td>+</td><td>$\mathfrak{so}(2,1)$</td></tr><tr><td>$\mathfrak{so}(3,1)$</td><td></td><td>$\mathfrak{sl}(2;\mathbb{C})$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(2,2)$</td><td></td><td>$\mathfrak{so}(2,1)$</td><td>+</td><td>$\mathfrak{so}(2,1)$</td></tr><tr><td>B_2</td><td>=</td><td>C_2</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5)$</td><td></td><td>$\mathfrak{sp}(2) = \mathfrak{usp}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4,1)$</td><td></td><td>$\mathfrak{sp}(1,1) = \mathfrak{usp}(2,2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3,2)$</td><td></td><td>$\mathfrak{sp}(4;R)$</td><td></td><td></td></tr><tr><td>D_3</td><td>=</td><td>A_3</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(6)$</td><td></td><td>$\mathfrak{su}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5,1)$</td><td></td><td>$\mathfrak{su}^*(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}^*(6)$</td><td></td><td>$\mathfrak{su}(3,1)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4,2)$</td><td></td><td>$\mathfrak{su}(2,2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3,3)$</td><td></td><td>$\mathfrak{sl}(4;\mathbb{R})$</td><td></td><td></td></tr></table>	A_1		B_1		C_1	$\mathfrak{su}(2)$		$\mathfrak{so}(3)$		$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$\mathfrak{su}(1,1) = \mathfrak{sl}(2;\mathbb{R})$		$\mathfrak{so}(2,1)$		$\mathfrak{sp}(2;\mathbb{R})$	D_2	=	A_1	+	A_1	$\mathfrak{so}(4)$	=	$\mathfrak{so}(3)$	+	$\mathfrak{so}(3)$	$\mathfrak{so}^*(4)$		$\mathfrak{so}(3)$	+	$\mathfrak{so}(2,1)$	$\mathfrak{so}(3,1)$		$\mathfrak{sl}(2;\mathbb{C})$			$\mathfrak{so}(2,2)$		$\mathfrak{so}(2,1)$	+	$\mathfrak{so}(2,1)$	B_2	=	C_2			$\mathfrak{so}(5)$		$\mathfrak{sp}(2) = \mathfrak{usp}(4)$			$\mathfrak{so}(4,1)$		$\mathfrak{sp}(1,1) = \mathfrak{usp}(2,2)$			$\mathfrak{so}(3,2)$		$\mathfrak{sp}(4;R)$			D_3	=	A_3			$\mathfrak{so}(6)$		$\mathfrak{su}(4)$			$\mathfrak{so}(5,1)$		$\mathfrak{su}^*(4)$			$\mathfrak{so}^*(6)$		$\mathfrak{su}(3,1)$			$\mathfrak{so}(4,2)$		$\mathfrak{su}(2,2)$			$\mathfrak{so}(3,3)$		$\mathfrak{sl}(4;\mathbb{R})$			D_2 $\mathfrak{so}(4)$ $\mathfrak{so}^*(4)$ $\mathfrak{so}(3,1)$ $\mathfrak{so}(2,2)$ B_2 $\mathfrak{so}(5)$ $\mathfrak{so}(4,1)$ $\mathfrak{so}(3,2)$ D_3 $\mathfrak{so}(6)$ $\mathfrak{so}(5,1)$ $\mathfrak{so}^*(6)$ $\mathfrak{so}(4,2)$ $\mathfrak{so}(3,3)$ = A_1 $\mathfrak{so}(3)$ $\mathfrak{so}(3)$ $\mathfrak{sl}(2;\mathbb{C})$ $\mathfrak{so}(2,1)$ C_2 $\mathfrak{sp}(2) = \mathfrak{usp}(4)$ $\mathfrak{sp}(1,1) = \mathfrak{usp}(2,2)$ $\mathfrak{sp}(4;R)$ A_3 $\mathfrak{su}(4)$ $\mathfrak{su}^*(4)$ $\mathfrak{su}(3,1)$ $\mathfrak{su}(2,2)$ $\mathfrak{sl}(4;\mathbb{R})$ + + + +
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gilmore-lie-groups_ TAB_019	196	■1.000	<pre>\begin{tabular}[t]{l l l l l} \hline \multirow[b]{2}{*}{Root space} & \multirow[b]{2}{*}{Class} & \multicolumn{2}{*}{Maximal compact subgroup} \\ \hline \text{rank(character)} & & Root space & Dimension \\ \hline G_2 & G_{2(-14)} & G_2 & 14 \\ & G_{2(+2)} & A_1 + A_1 & 6 \\ F_4 & F_{4(-52)} & F_4 & 52 \\ & F_{4(-20)} & B_4 & 36 \\ & F_{4(+4)} & C_3 + A_1 & 24 \\ E_6 & E_{6(-78)} & E_6 & 78 \\ & E_{6(-26)} & F_4 & 52 \\ & E_{6(-14)} & D_5 + D_1 & 46 \\ & E_{6(+2)} & A_5 + A_1 & 38 \\ & E_{6(+6)} & C_4 & 36 \\ E_7 & E_{7(-133)} & E_7 & 133 \\ & E_{7(-25)} & E_6 + D_1 & 79 \\ & E_{7(-5)} & D_6 + A_1 & 69 \\ & E_{7(+7)} & A_7 & 63 \\ E_8 & E_{8(-248)} & E_8 & 248 \\ & E_{8(-24)} & E_7 + A_1 & 136 \\ & E_{8(+8)} & D_8 & 120 \\ \hline \hline \end{tabular}</pre>	(not rendered)	<table><thead><tr><th colspan="2"></th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Class</th><th>Root space</th><th>Dimension</th></tr></thead><tbody><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></tbody></table> completeness we list the real forms for the exceptional Li			Maximal compact subgroup		Root space	Class	Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ TAB_020	199	■1.000	<pre>\begin{tabular}[t]{l l l l l} \hline Quadrant & Operator & Edge \\ \hline I & J_+ & n_2 = 0 \\ \hline I & J_- & n_1 = 0 \\ \hline III & J_+ & n_1 = -1 \\ \hline III & J_- & n_2 = -1 \\ \hline \end{tabular}</pre>	<table><thead><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr></thead><tbody><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></tbody></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$	<table><thead><tr><th>III</th><th>J_+</th><th>$n_1 = -1$</th></tr><tr><th>III</th><th>J_-</th><th>$n_2 = -1$</th></tr></thead></table> elements for the shift operators $K_{\pm}$ in $\mathfrak{su}(l)$ $i\sqrt{(j \mp m)(j \pm m + 1)} = \sqrt{\left(m \pm \frac{1}{2}\right)^2 -}$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$																																											
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gilmore-lie-groups_ TAB_021	206	0.877	$\begin{tabular}{t l l l l } \hline \text{Root space} \& \text{Quotient } (G^{\prime} / H) \& \text{Dimension } (P) \& \text{Rank } P \\ \hline A_{p+q-1} \& SU(p,q)/S[U(p) \otimes U(q)] \& 2pq \& \min(p,q) \\ A_{n-1} \& SL(n;\mathbb{R})/SO(n) \& \frac{1}{2}(n+2)(n-1) \& n-1 \\ A_{2n-1} \& SU^*(2n)/U\mathfrak{Sp}(2n) \& (2n+1)(n-1) \& n-1 \\ B_{p+q} \& SO(p,q)/SO(p) \otimes SO(q) \& pq \& \min(p,q) \\ D_{p+q} \& SO(p,q)/SO(p) \otimes SO(q) \& pq \& \min(p,q) \\ D_n \& SO^*(2n)/U(n) \& n(n-1) \& [n/2] \\ C_{p+q} \& USp(2p,2q)/USp(2p) \otimes USp(2q) \& 4pq \& \min(p,q) \\ C_n \& Sp(2n;\mathbb{R})/U(n) \& n(n+1) \& n \\ \hline \end{tabular}$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\mathfrak{Sp}(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\mathfrak{Sp}(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p) \otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p) \otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n	<table><tr><td>A_{2n-1}</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>B_{p+q}</td><td>$SO(p,q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p) \otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p) \otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																		
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C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n																																																																																
gilmore-lie-groups_ TAB_022	207	0.979	$\begin{tabular}{t l l l l l l } \hline \text{Root space} \& (G^{\prime} / H) \& (\operatorname{Dim} G^{\prime}) \& (\operatorname{Dim} H) \& (\operatorname{Dim} P) \& \text{Rank } P \\ \hline G_2 \& (G_{2(+2)}) \& 14 \& 6 \& 8 \& 2 \\ F_4 \& (F_{4(-20)}) \& 52 \& 36 \& 16 \& 1 \\ & (F_{4(+4)}) \& 52 \& 24 \& 28 \& 4 \\ E_6 \& (E_{6(-26)}) \& 78 \& 52 \& 26 \& 2 \\ & (E_{6(-14)}) \& 78 \& 46 \& 32 \& 2 \\ & (E_{6(+2)}) \& 78 \& 38 \& 40 \& 4 \\ & (E_{6(+6)}) \& 78 \& 36 \& 42 \& 6 \\ E_7 \& (E_{7(-25)}) \& 133 \& 79 \& 54 \& 3 \\ & (E_{7(-5)}) \& 133 \& 69 \& 64 \& 4 \\ & (E_{7(+7)}) \& 133 \& 63 \& 70 \& 7 \\ E_8 \& (E_{8(-24)}) \& 248 \& 136 \& 112 \& 4 \\ & (E_{8(+8)}) \& 248 \& 120 \& 128 \& 8 \\ \hline \end{tabular}$	(not rendered)	<table><tr><th>Root space</th><th>G'/H</th><th>Dim G'</th><th>Dim H</th><th>Dim P</th><th>Rank P</th></tr><tr><td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr><tr><td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr><tr><td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr><tr><td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr><tr><td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr><tr><td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr><tr><td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr><tr><td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr><tr><td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr><tr><td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr><tr><td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr><tr><td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr></table> subgroup in G'. To each there is a compact real form under $G'/H \rightarrow G/H$. For	Root space	G'/H	Dim G'	Dim H	Dim P	Rank P	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
Root space	G'/H	Dim G'	Dim H	Dim P	Rank P																																																																														
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gilmore-lie-groups_ TAB_023	212	■ 1.000	$\begin{array}{ l l l l } \hline \text{Operation} & \sigma_1 & \sigma_2 & \sigma_3 \\ \hline T_1 = \text{block matrix decomposition} & -1 & -1 & +1 \\ T_2 = \text{complex conjugation} & -1 & +1 & -1 \\ T_3 = T_1 T_2 & +1 & -1 & -1 \\ \hline \end{array}$	$\begin{array}{ l l l l } \hline \text{Operation} & \sigma_1 & \sigma_2 & \sigma_3 \\ \hline T_1 = \text{block matrix decomposition} & -1 & -1 & +1 \\ T_2 = \text{complex conjugation} & -1 & +1 & -1 \\ T_3 = T_1 T_2 & +1 & -1 & -1 \\ \hline \end{array}$	$\begin{array}{ l l l l } \hline T_1 = \text{block matrix decomposition} & -1 & -1 & +1 \\ T_2 = \text{complex conjugation} & -1 & +1 & -1 \\ T_3 = T_1 T_2 & +1 & -1 & -1 \\ \hline \end{array}$ as $\mathfrak{g}_{+,+} = 0$, $\mathfrak{g}_{+,-} = i\sigma_3$, $\mathfrak{g}_{-,+} = i\sigma_2$, $\mathfrak{g}_{-,-} = i\sigma_1$. Note tha
gilmore-lie-groups_ TAB_024	232	■ 1.000	$\begin{array}{ l l l l } \hline \text{Bessel function} & \rightarrow & \text{harmonic oscillator eigenfunction} \\ \text{Bessel equation} & \rightarrow & \text{Schrödinger equation for harmonic oscillator} \\ \hline \end{array}$	$\begin{array}{ l l l l } \hline \text{Bessel function} & \rightarrow & \text{harmonic oscillator eigenfunction} \\ \text{Bessel equation} & \rightarrow & \text{Schrödinger equation for harmonic oscillator} \\ \hline \end{array}$	Contract the Lie algebra $\mathfrak{su}(2)$ spanned by $\sigma_3, \sigma_{\pm}$ ($[\sigma_3, \sigma_{\pm}] = \pm\sigma_{\pm}$, $[\sigma_+, \sigma_-] = 2\sigma_3$) with respect to the subalgebra J_-. Use a simple Inönü–Wigner contraction to show
gilmore-lie-groups_ TAB_025	239	■ 0.424	$\begin{array}{ l l l l } \hline \text{Equation} & A & B & C \\ \hline \text{Klein-Gordon} & -l(l+1) + (e^2/\hbar c)^2 & 2Ee^2/(\hbar c)^2 & [E^2 - (mc^2)^2]/(\hbar c)^2 \\ \text{Schrödinger} & -l(l+1) & 2me^2/\hbar^2 & 2mW/\hbar^2 \\ \hline \end{array}$	$\begin{array}{ l l l l } \hline \text{Equation} & A & B & C \\ \hline \text{Klein-Gordon} & -l(l+1) + (e^2/\hbar c)^2 & 2Ee^2/(\hbar c)^2 & [E^2 - (mc^2)^2]/(\hbar c)^2 \\ \text{Schrödinger} & -l(l+1) & 2me^2/\hbar^2 & 2mW/\hbar^2 \\ \hline \end{array}$	$\begin{array}{ l l l l } \hline \text{Schrödinger} & -l(l+1) & 2me^2/\hbar^2 & 2mW/\hbar^2 \\ \hline \end{array}$ There is a standard procedure for solving simple ordinary differential equations of the type presented in Eq. (14.9). This is the Frobenius method. The steps involved in this method, and the result of each step, are summarized in Table 14.1.

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gilmore-lie-groups_ TAB_026	240	0.655	<pre> \begin{tabular}[t]{l l l } \hline & Procedure & Result \\ \hline 1 & Locate singularities & $0, \infty$ \\ 2 & Determine analytic behavior at singular points & $r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma-1)+A=0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2+C=0$ \\ 3 & Keep only $\mathcal{L}^2$ solutions & $\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$ \\ 4 & Look for solutions with proper asymptotic behavior & $R = r^\gamma e^{\lambda r} f(r)$ \\ 5 & Construct differential equation for $f(r)$ & $[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$ \\ 6 & Construct recursion relation & $f_{j+1} = -\frac{2\lambda(j+\gamma)+B}{j(j+1)+2\gamma(j+1)} f_j$ \\ 7 & Look at asymptotic behavior & $f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$) \\ 8 & Construct quantization condition & $2\lambda(n+\gamma)+B=0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$ \\ 9 & Construct explicit solutions & $E = \frac{mc^2}{\sqrt{1+(\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l+\frac{1}{2})^2 - \alpha^2}, N = n+l+1$ \\ \hline \hline </pre>	(not rendered)	Procedure Result 1 Locate singularities $0, \infty$ 2 Determine analytic behavior at singular points $r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma-1)+A=0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2+C=0$ 3 Keep only $\mathcal{L}^2$ solutions $\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$ 4 Look for solutions with proper asymptotic behavior $R = r^\gamma e^{\lambda r} f(r)$ 5 Construct differential equation for $f(r)$ $[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$ 6 Construct recursion relation $f_{j+1} = -\frac{2\lambda(j+\gamma)+B}{j(j+1)+2\gamma(j+1)} f_j$ 7 Look at asymptotic behavior $f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$) 8 Construct quantization condition $2\lambda(n+\gamma)+B=0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$ 9 Construct explicit solutions $E = \frac{mc^2}{\sqrt{1+(\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l+\frac{1}{2})^2 - \alpha^2}, N = n+l+1$
This is a dimensionless ratio of three physical constants that are fundamental in three “different” areas of physics: e (electromagnetism), $\hbar$ (quantum mechanics),					

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gilmore-lie-groups_ TAB_027	244	■0.998	$\begin{tabular}{t}{ l l l l l }\hline \backslash \hline \backslash (m\backslash) \& \backslash (l=0\backslash) \& \backslash (l=1\backslash) \& \backslash (l=2\backslash) \& \backslash (l=3\backslash) \backslash \backslash \hline 0 \& \backslash (\sqrt{\frac{1}{4\pi}}\backslash) \& \backslash (\sqrt{\frac{3}{4\pi}}\backslash) \backslash \cos \theta \& \backslash (\sqrt{\frac{5}{16\pi}}\backslash) \backslash \left(3 \cos^2 \theta - 1 \right) \& \backslash (\sqrt{\frac{7}{16\pi}}\backslash) \backslash \left(5 \cos^3 \theta - 3 \cos \theta \right) \backslash \hline \pm 1 \& \backslash (\sqrt{\frac{3}{8\pi}}\backslash) \backslash \sin \theta e^{\pm i \phi} \& \backslash (\sqrt{\frac{15}{8\pi}}\backslash) \backslash \cos \theta \sin \theta e^{\pm i \phi} \& \backslash (\sqrt{\frac{21}{64\pi}}\backslash) \backslash \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi} \& \backslash (\sqrt{\frac{35}{64\pi}}\backslash) \backslash \sin^3 \theta e^{\pm 3 i \phi} \backslash \hline \pm 2 \& \backslash (\sqrt{\frac{15}{32\pi}}\backslash) \backslash \sin^2 \theta e^{\pm 2 i \phi} \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \backslash \sin^2 \theta \cos \theta e^{\pm 2 i \phi} \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \backslash \sin^2 \theta \cos \theta e^{\pm 2 i \phi} \& \backslash (\sqrt{\frac{105}{32\pi}}\backslash) \backslash \sin^2 \theta \cos \theta e^{\pm 2 i \phi} \backslash \hline \pm 3 \& \backslash (\sqrt{\frac{35}{64\pi}}\backslash) \backslash \sin^3 \theta e^{\pm 3 i \phi} \& \backslash (\sqrt{\frac{35}{64\pi}}\backslash) \backslash \sin^3 \theta e^{\pm 3 i \phi} \& \backslash (\sqrt{\frac{35}{64\pi}}\backslash) \backslash \sin^3 \theta e^{\pm 3 i \phi} \& \backslash (\sqrt{\frac{35}{64\pi}}\backslash) \backslash \sin^3 \theta e^{\pm 3 i \phi} \backslash \hline \end{tabular}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} \left(3 \cos^2 \theta - 1\right)$</td><td>$\sqrt{\frac{7}{16\pi}} \left(5 \cos^3 \theta - 3 \cos \theta\right)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta \left(5 \cos^2 \theta - 1\right) e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2 i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2 i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3 i \phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} \left(3 \cos^2 \theta - 1\right)$	$\sqrt{\frac{7}{16\pi}} \left(5 \cos^3 \theta - 3 \cos \theta\right)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta \left(5 \cos^2 \theta - 1\right) e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2 i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2 i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3 i \phi}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} \left(3 \cos^2 \theta - 1\right)$</td><td>$\sqrt{\frac{7}{16\pi}} \left(5 \cos^3 \theta - 3 \cos \theta\right)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta \left(5 \cos^2 \theta - 1\right) e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2 i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2 i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3 i \phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} \left(3 \cos^2 \theta - 1\right)$	$\sqrt{\frac{7}{16\pi}} \left(5 \cos^3 \theta - 3 \cos \theta\right)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta \left(5 \cos^2 \theta - 1\right) e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2 i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2 i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3 i \phi}$				
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gilmore-lie-groups_ TAB_028	258	■0.721	$\begin{tabular}{t}{ l l l l }\hline H \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \& \backslash (\frac{p^2}{2m} - \frac{\alpha}{r}) \backslash \hline A_4 \& \mathbf{r} \cdot \mathbf{p} - i \& \mathbf{r} \cdot \mathbf{p} - i \& \mathbf{r} \cdot \mathbf{p} - i \backslash \hline \Gamma_4 \& \frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r) \& \frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r) \& \frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}) \backslash \hline \Gamma_5 \& \frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r) \& \frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r) \& \frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}) \backslash \hline \Theta \& r (H_S - W) \& r (H_S - W) \& r \left\{ \left(H_{KG} + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\} \backslash \hline A \& \frac{1}{2m} \& \frac{1}{2m} \& \frac{1}{2m} \backslash \hline B \& -W \& -W \& m^2 - E^2 \backslash \hline C \& -\alpha \& -\alpha \& -2\alpha E \backslash \hline \end{tabular}$	<table><tr><th>H</th><th>$\frac{p^2}{2m} - \frac{\alpha}{r}$</th><th>$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r)$</td><td>$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r} \right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r)$</td><td>$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r} \right)$</td></tr><tr><td>$\Theta$</td><td>$r (H_S - W)$</td><td>$r \left\{ \left(H_{KG} + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\}$</td></tr><tr><td></td><td>$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$</td><td>$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$</td></tr><tr><td>A</td><td>$1/2m$</td><td>1</td></tr><tr><td>B</td><td>$-W$</td><td>$m^2 - E^2$</td></tr><tr><td>C</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$	A_4	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	Γ_4	$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r} \right)$	Γ_5	$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r} \right)$	Θ	$r (H_S - W)$	$r \left\{ \left(H_{KG} + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\}$		$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$	$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$	A	$1/2m$	1	B	$-W$	$m^2 - E^2$	C	$-\alpha$	$-2\alpha E$	<table><tr><th>H</th><th>$\frac{p^2}{2m} - \frac{\alpha}{r}$</th><th>$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r)$</td><td>$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r} \right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r)$</td><td>$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r} \right)$</td></tr><tr><td>$\Theta$</td><td>$r (H_S - W)$</td><td>$r \left\{ \left(H_{KG} + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\}$</td></tr><tr><td></td><td>$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$</td><td>$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$</td></tr><tr><td>A</td><td>$1/2m$</td><td>1</td></tr><tr><td>B</td><td>$-W$</td><td>$m^2 - E^2$</td></tr><tr><td>C</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table> In the event a magnetic field $\mathbf{B}$ is present, the momentum operators $\mathbf{p}$ should be replaced by $\pi = \mathbf{p} - \frac{q}{c} \mathbf{A}$. Under this condition the operators still close under	H	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$	A_4	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	Γ_4	$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r} \right)$	Γ_5	$\frac{1}{2} (r \mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2} \left(r \mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r} \right)$	Θ	$r (H_S - W)$	$r \left\{ \left(H_{KG} + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\}$		$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$	$A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$	A	$1/2m$	1	B	$-W$	$m^2 - E^2$	C	$-\alpha$	$-2\alpha E$
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gilmore-lie-groups_ TAB_030	269	■ 1.000	$\begin{tabular}[t]{ l l l l l l l l } \hline \text{Mercury} & \text{Venus} & \text{Earth} & \text{Mars} & \text{Jupiter} & \text{Saturn} & \text{Uranus} & \text{Neptune} \\ \hline 0.39 & 0.72 & 1.00 & 1.52 & 5.20 & 9.54 & 19.18 & 30.06 \\ \hline \end{tabular}$	<table><tr><th>Mercury</th><th>Venus</th><th>Earth</th><th>Mars</th><th>Jupiter</th><th>Saturn</th><th>Uranus</th><th>Neptune</th></tr><tr><td>0.39</td><td>0.72</td><td>1.00</td><td>1.52</td><td>5.20</td><td>9.54</td><td>19.18</td><td>30.06</td></tr></table>	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	0.39	0.72	1.00	1.52	5.20	9.54	19.18	30.06	<table><tr><th>Mercury</th><th>Venus</th><th>Earth</th><th>Mars</th><th>Jupiter</th><th>Saturn</th><th>Uranus</th><th>Neptune</th></tr><tr><td>0.39</td><td>0.72</td><td>1.00</td><td>1.52</td><td>5.20</td><td>9.54</td><td>19.18</td><td>30.06</td></tr></table>	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	0.39	0.72	1.00	1.52	5.20	9.54	19.18	30.06												
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gilmore-lie-groups_ TAB_031	274	■ 1.000	$\begin{tabular}[t]{ l l l l } \hline \text{Time period} & \text{Approach} & \text{Strengths} & \text{Weaknesses} \\ \hline \text{Nineteenth century} & \text{Manifestly covariant} & \text{Fields have elegant transformation properties} & \text{Many fields represent nonphysical states} \\ \hline \text{Twentieth century} & \text{Hilbert space} & \text{All linear superpositions represent physical states} & \text{Transformation properties are complicated} \\ \hline \end{tabular}$	<table><tr><th>Time period</th><th>Approach</th><th>Strengths</th><th>Weaknesses</th></tr><tr><td>Nineteenth century</td><td>Manifestly covariant</td><td>Fields have elegant transformation properties</td><td>Many fields represent nonphysical states</td></tr><tr><td>Twentieth century</td><td>Hilbert space</td><td>All linear superpositions represent physical states</td><td>Transformation properties are complicated</td></tr></table>	Time period	Approach	Strengths	Weaknesses	Nineteenth century	Manifestly covariant	Fields have elegant transformation properties	Many fields represent nonphysical states	Twentieth century	Hilbert space	All linear superpositions represent physical states	Transformation properties are complicated	<table><tr><th>Time period</th><th>Approach</th><th>Strengths</th><th>Weaknesses</th></tr><tr><td>Nineteenth century</td><td>Manifestly covariant</td><td>Fields have elegant transformation properties</td><td>Many fields represent nonphysical states</td></tr><tr><td>Twentieth century</td><td>Hilbert space</td><td>All linear superpositions represent physical states</td><td>Transformation properties are complicated</td></tr></table>	Time period	Approach	Strengths	Weaknesses	Nineteenth century	Manifestly covariant	Fields have elegant transformation properties	Many fields represent nonphysical states	Twentieth century	Hilbert space	All linear superpositions represent physical states	Transformation properties are complicated																				
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gilmore-lie-groups_ TAB_032	280	■ 0.815	$\begin{tabular}[t]{ l l l } \hline & k\rangle & \begin{matrix} j & j' \\ \mu & \mu' \end{matrix} \\ \hline \{I, a\} & k\rangle e^{ik\cdot a} & \begin{matrix} j & j' \\ v & v' \end{matrix} \end{matrix} \Bigg\rangle_{vv';\mu\mu'}$	<table><tr><td></td><td>$k\rangle$</td><td>$\begin{matrix} j & j' \\ \mu & \mu' \end{matrix} \Bigg\rangle$</td></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}$</td></tr><tr><td>$\{ \Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}^{jj'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\begin{matrix} j & j' \\ \mu & \mu' \end{matrix} \Bigg\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}$	$\{ \Lambda, 0\}$	$ \Lambda k\rangle$	$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}^{jj'}(\Lambda)$	<table><tr><td></td><td>$k\rangle$</td><td>$\begin{matrix} j & j' \\ \mu & \mu' \end{matrix} \Bigg\rangle$</td></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}$</td></tr><tr><td>$\{ \Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}^{jj'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\begin{matrix} j & j' \\ \mu & \mu' \end{matrix} \Bigg\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}$	$\{ \Lambda, 0\}$	$ \Lambda k\rangle$	$\begin{matrix} j & j' \\ v & v' \end{matrix} \Bigg\rangle_{vv';\mu\mu'}^{jj'}(\Lambda)$																										
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Identifier	Page	Conf.	LaTeX source	Rendered	Image												
gilmore-lie-groups_ TAB_033	290	■ 1.000	<pre>\begin{tabular}[t]{ l l l } \hline & Electric & Magnetic \\ \hline Charge density & \(\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)\) & \(\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)\) \\ \hline Current density & \(\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}\) & \(\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}\) \\ \hline Conservation law & \(\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0\) & \(\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0\) \\ \hline \end{tabular}</pre>	<table><tr><th></th><th>Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$	Charge density $\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$ $\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$ Current density $\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$ $\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$ Conservation law $\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$ $\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$ The conservation equations enforce the conditions of charge conservation (both elec-
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gilmore-lie-groups_ TAB_035	316	■1.000	$\begin{tabular}{t}{ l l l } \hline 1830 & \text{Galois} & \text{solve algebraic equations} \\ \hline 1874 & \text{Lie} & \text{solve differential equations} \\ \hline 1969 & \text{Risch} & \text{integrate in closed form} \\ \hline ? & - & \text{solve determining equations for } \xi, \eta \\ \hline ? & - & \text{solve determining equations for } R, S, T \\ \hline ? & - & \text{solve } F(R, -, T) = 0 \text{ for } R. \\ \hline \end{tabular}$	<table><tr><td>1830</td><td>Galois</td><td>solve algebraic equations</td></tr><tr><td>1874</td><td>Lie</td><td>solve differential equations</td></tr><tr><td>1969</td><td>Risch</td><td>integrate in closed form</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for ξ, η</td></tr><tr><td>?</td><td>-</td><td>solve determining equations for R, S, T</td></tr><tr><td>?</td><td>-</td><td>solve $F(R, -, T) = 0$ for R.</td></tr></table>	1830	Galois	solve algebraic equations	1874	Lie	solve differential equations	1969	Risch	integrate in closed form	?	-	solve determining equations for ξ, η	?	-	solve determining equations for R, S, T	?	-	solve $F(R, -, T) = 0$ for R .	1830 Galois solve algebraic equations 1874 Lie solve differential equations 1969 Risch integrate in closed form ? – solve determining equations for ξ, η ? – solve determining equations for R, S, T ? – solve $F(R, -, T) = 0$ for R. It is clear that additional algorithms are nossible and desirable.												
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gilmore-lie-groups_ TAB_038	320	■1.000	$\begin{tabular}{t}{ l l l } \hline \text{Symmetry} & \text{Conserved quantity} \\ \hline \text{Space displacements} & \text{momentum} \\ \hline \text{Time displacements} & \text{energy} \\ \hline \text{Space-time displacements} & \text{four-momentum} \\ \hline \text{Rotations} & \text{angular momentum} \\ \hline \end{tabular}$	<table><tr><th>Symmetry</th><th>Conserved quantity</th></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum	<table><tr><th>Rotations</th><th>angular momentum</th></tr></table>	Rotations	angular momentum																		
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